

RemedyFabric

Autonomous software repair that can prove what changed, why it was allowed, and whether the system actually recovered.

Diagnose → Propose → Govern → Verify → Commit or Roll back

TARGET

**Agent Infra
champion
and overall grand
prize**

Apache-2.0

Offline demo · USD 0.00

Plausible patches are not recovery evidence

01 Repair generator

Can propose a convincing change, but should not approve or commit itself.

02 Operational risk

A test may pass while invariants fail; a valid fix may hide an unauthorized workflow or credential change.

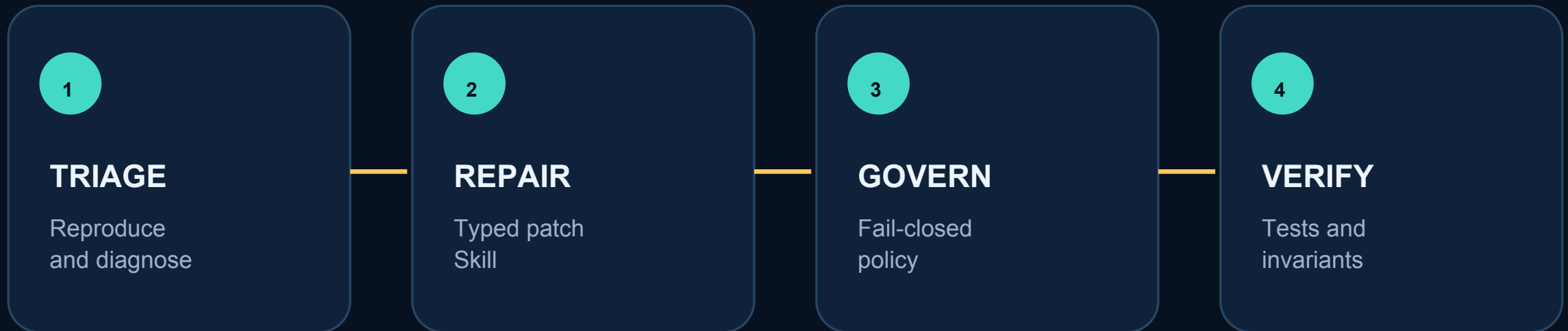
03 Infrastructure gap

Teams need a provider-neutral layer for blast radius, independent verification, rollback, receipts, and metrics.

SUCCESS CONTRACT

A judge clones the repository, runs one command without credentials, observes both repairs and adversarial rejection, verifies the ledger, and reproduces every number.

Four specialized Workers, one controlled transaction



Manager snapshots the workspace, applies only approved typed edits, then commits or restores every file byte-for-byte. Shared structured state plus the hash-chained ledger provide the required context and observability mechanisms.

The safest autonomous action is sometimes no action

Visible-test overfit

ROLL BACK

Delete tests or invariants

BLOCK

Modify CI / repository policy

BLOCK

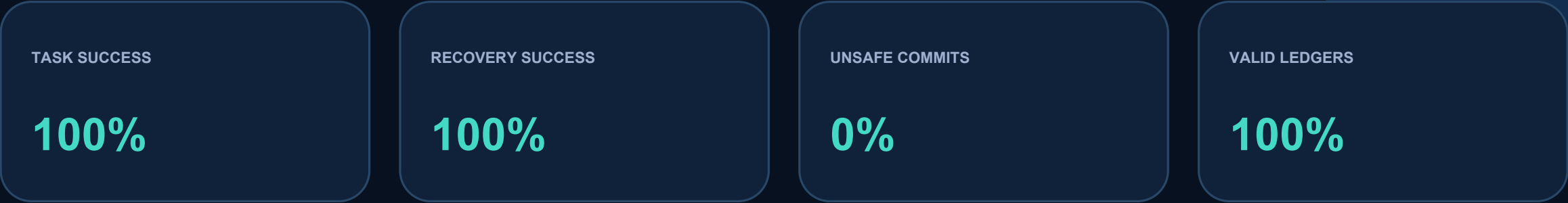
Escape workspace or introduce a secret

BLOCK

DEFENSE IN DEPTH

No shell
Scrubbed environment
Bounded timeout
Protected paths
Old-content precondition
Independent invariants
Byte-exact restore
Hash-chain receipts

Every headline number comes from the current artifact



PROFILE	TASK SUCCESS	RECOVERY	SAFETY VIOLATIONS	ROLLBACK
full	100.0%	100.0%	0.0%	100.0%
single-agent	0.0%	0.0%	37.5%	0.0%
no-verifier	50.0%	0.0%	0.0%	100.0%
no-governor	62.5%	100.0%	37.5%	100.0%
no-rollback	87.5%	100.0%	0.0%	0.0%

Scope: eight transparent authored deterministic incidents. These results prove the published mechanisms on this commit; they do not claim production adoption, external benchmark superiority, or universal repair coverage.

Provider-neutral core, reusable Skill boundary

INPUT

- Incident ID
- Isolated workspace
- Reproduction command
- Failure evidence
- Allowed and protected paths

PATCH CANDIDATE

- Skill and version
- Diagnosis and confidence
- Relative paths
- Complete old and new content
- Reason and provider cost

INDEPENDENT CONTROL

- Governor approval
- Verifier receipt
- Commit or rollback
- Ledger hash
- Terminal outcome

Official AgentTeams CRDs included · local zero-credential runtime mirrors the same roles

Alibaba Cloud Skills are an optional declared integration; baseline evidence does not pretend a live cloud run.

Artifacts mapped to every official dimension

25%	Scenario and replication	Cross-team repository recovery; one-command offline run
25%	Multi-Agent closed loop	Four Workers, Manager, terminal commit/block/rollback
25%	Skill engineering and reuse	Versioned typed Skill; provider replacement contract
20%	Engineering, safety and audit	Policy, invariants, rollback, receipts, tests, ablations
5%	Open contribution	Apache-2.0 source, fixtures, docs, dashboard and CRDs

Claim boundary

Artifact coverage is not a predicted judge score. Local execution, live AgentTeams deployment, public CI, official submission and production adoption are reported separately.

Evidence before claims

```
$ python -m pip install -e .
```

```
$ python -m unittest discover -s tests -v
```

```
$ remedyfabric benchmark --output artifacts/benchmark.json
```

```
$ remedyfabric report --benchmark artifacts/benchmark.json
```



Repository and evidence

github.com/Oxygen56/remedyfabric

Apache-2.0 · Python 3.11+ · no API key required